Refining Regional Marine Gravity Field Through Fusion of Satellite Altimetry and Shipborne Data Using Radial Basis Functions

Wenqi Lin, Jiancheng Li, Xinyu Xu, Yongqi Zhao, and Mao Zhou

***Abstract*—Satellite altimetry-derived and shipborne gravity anomalies exhibit complementary spectral characteristics and spatial resolutions. This study develops an iterative radial basis function (RBF) framework to refine regional marine gravity fields through synergistic fusion of these datasets. Using the Poisson kernel RBF with the spatial and spectral localization properties, we first construct an initial regional gravity model. This model then detects/removes outliers and corrects systematic errors in shipborne data. The updated shipborne measurements are reincorporated into an improved RBF model, with the process iterating until the model residuals converge. Evaluated in the Ogasawara Islands and central-eastern Philippine Sea regions, our iterative RBF approach demonstrates significant accuracy gains over least-squares collocation (LSC) and noniterative RBF methods. The accuracy improvements are 53.14% (versus noniterative RBF), 28.59% (versus DTU17), and 16.68% (versus SS 32.1) in the Ogasawara region, and 16.79% (versus noniterative RBF), 40.23% (versus DTU17), and 40.56% (versus SS 32.1) in the Philippine Sea region. Power spectrum and coherence analyses confirm the RBF model enhances signal content within the 817 km wavelength range, reducing the effective resolution from ~17 to <10 km. This demonstrates the framework's capability to resolve finer-scale ocean gravity structures by combining the satellite altimetry-derived and shipborne gravity anomalies critical for geophysical and oceanographic applications.**

***Index Terms*—Radial basis functions (RBFs), regional marine gravity field refinement, satellite altimetry, shipborne gravity.**

I. Introduction

APPROXIMATELY 71% of the Earth's surface is covered by oceans. A refined marine gravity field model is indispensable for the effective exploration and development of marine resources. Moreover, high-accuracy marine gravity data are essential for numerous applications, such as inertial navigation, seafloor topography inversion, and sea-level variability analysis [1], [2], [3]. Presently, satellite altimetry and shipborne gravity surveys are critical technologies for marine gravity data acquisition. Satellite altimetry provides globally distributed, high-accuracy sea surface height measurements, enabling accurate inversion of medium-to-long wavelength gravity field signals. However, its capability to reveal short-wavelength gravity signals is limited. Shipborne gravity data exhibit high accuracy and are rich in short-wavelength signals, but they are expensive to obtain, regionally limited, and sparsely distributed. By combining the advantages of both data types, it is possible to maximize the extraction of gravity field information [4], [5], thereby ensuring the construction of high-accuracy, high-resolution regional gravity field models [6].

Global gravity field models are typically expressed in terms of spherical harmonics [7], but the inherent global characteristic of spherical harmonics limits their suitability for representing regional gravity fields [8]. Classical numerical integration methods, such as Stokes and Hotine, face difficulties in fusing multisource gravity data [9]. The least-squares collocation (LSC) [10], [11], as a representative statistical method, requires the determination of variance-covariance functions for multisource gravity data, thereby increasing the complexity of its application. In recent years, radial basis functions (RBFs) have been widely applied in regional gravity field modeling due to their compact support properties [4], [12]. There are various types of RBFs, including point mass kernel [13], [14], radial multipole [15], [16], Poisson kernel [17], [18], Poisson wavelet [19], [20], [21], Blackman function [22], [23], [24], Shannon function [8], [24], [25], and cubic polynomial (CuP) kernel [26], etc. Several RBFs have been successfully applied to construct regional gravity field models. For instance, the radial multipole was utilized to construct a geoid model for the Skagerrak region of Scandinavia, achieving an accuracy of 5 cm [15]. The Poisson wavelet was applied to construct the NLGEO2007 height datum for the Netherlands [27], with an rms error of 0.6 cm relative to the official NLGEO2004 datum. The RBF method was also employed in the modeling and downward continuation of regional airborne gravity data [28], demonstrating its significance and contribution to high-resolution regional gravity field modeling. A combination of Shannon and CuP functions was used to compute a high-resolution geoid model for Colorado, achieving an rms error of 1.3 cm at the GSVS17 benchmark [29]. In addition, RBFs have been applied in multisource data fusion. For example, Shannon and Blackman functions were used to fuse satellite gravity, satellite altimetry, airborne, shipborne, and terrestrial gravity data to construct a gravimetric quasi-geoid model for Northern Germany [30], with cross-validation indicating a standard deviation below 5 cm.

Received 16 July 2025; revised 6 November 2025; accepted 28 November 2025. Date of publication 1 December 2025; date of current version 20 February 2026. This work was supported by the National Natural Science Foundation of China under Grant 42192533, Grant 42388102, Grant 42574007, Grant 42074019, and Grant 42404005. *(Corresponding author: Jiancheng Li.)*

The authors are with the MOE Key Laboratory of Geospace Environment and Geodesy, School of Geodesy and Geomatics, Wuhan University, Wuhan 430079, China (e-mail: wenqilin@whu.edu.cn; jcli@whu.edu.cn; xyxu@sgg.whu.edu.cn; yqzhao@whu.edu.cn; maozhou@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3639157

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

The Poisson wavelet was employed to fuse satellite altimetry and airborne gravity data to compute a gravimetric quasi-geoid in the northwestern South China Sea [31], which was subsequently used to calculate the mean dynamic topography (MDT), showing that the MDT error can be reduced by several centimeters. The Shannon function, employed within a two-scale RBF method, was used to fuse terrestrial and airborne gravity data [32], constructing a geoid model for the mountainous regions of California and Oregon, with a standard deviation of 9.8 cm compared to GPS/leveling data. The aforementioned studies affirm the applicability of the RBF method in constructing regional gravity field models. However, research specifically focused on refining regional marine gravity field models using RBFs remains sparse. This study proposes an iterative framework based on RBFs to fuse satellite altimetry and shipborne gravity data, directly incorporating shipborne data processing into the model construction. Experiments to refine the regional gravity field were conducted in the Ogasawara Islands and the central-eastern Philippine Sea. The results demonstrate the feasibility of an iterative RBF framework for refining the regional marine gravity field by fusing satellite altimetry and shipborne data.

The remaining sections of the article are organized as follows. Section II elaborates the methods, including the Poisson kernel RBF, the Poisson kernel RBF method, and the iterative RBF framework. Section III presents the study regions and data. Section IV provides the results and analysis. Section V concludes this article.

II. Method

A. Poisson Kernel

The Poisson kernel RBF Ψ_{pk} exhibits compact support properties in both the spatial and spectral domains, making it advantageous for regional gravity field modeling. It possesses an analytical form, and its computation is relatively simple. Under suitable depth conditions, the differences in modeling accuracy among various RBFs are minimal [17]. Therefore, Ψ_{pk} is selected as the basis function for modeling. The series expansion form of the Ψ_{pk} is as follows [17], [27]:

$$\Psi_{\mathrm{pk}}(\boldsymbol{z},\boldsymbol{y}) = \sum_{n=0}^{\infty}\left(\frac{|\boldsymbol{y}|}{|\boldsymbol{z}|}\right)^{n+1}(2n+1)\,P_n\left(\hat{\boldsymbol{z}}^{\boldsymbol{T}}\hat{\boldsymbol{y}}\right). \tag{1}$$

In (1), $\Psi_{\mathrm{pk}}(\boldsymbol{z},\boldsymbol{y})$ depends solely on the function location $\boldsymbol{y}$ and observation location $\boldsymbol{z}$; $|\cdot|$ is the radial distance from the geocenter to the function's location or observation point; Here, $\hat{z}$ represents the unit vector in the z-direction, while $\hat{y}$ represents the unit vector in the $\boldsymbol{y}$-direction; $P_n(\hat{\boldsymbol{z}}^{\boldsymbol{T}}\hat{\boldsymbol{y}})$ is the Legendre function of degree n. The RBF is situated beneath the *Bjerhammar* sphere, and its depth is defined as $d = R-|\boldsymbol{y}|$, where R is the radius of the *Bjerhammar* sphere.

The analytical form of the Ψ_{pk} is

$$\Psi_{\mathrm{pk}}(\boldsymbol{z},\boldsymbol{y}) = |\boldsymbol{y}|\frac{|\boldsymbol{z}|^2-|\boldsymbol{y}|^2}{|\boldsymbol{z}-\boldsymbol{y}|^3}. \tag{2}$$

The properties of the Poisson kernel RBF Ψ_{pk} in both spatial and spectral domains are discussed on the unit sphere. Fig. 1(a) illustrates the function values of Ψ_{pk} on the unit sphere at depth $d = 0.10$, with the function center located at the North Pole. Ψ_{pk} attains its maximum at the North Pole, and rapidly decays to zero within the Arctic Circle. Ψ_{pk} is more sensitive to gravity observations near the function center, while distant gravity observations contribute little or even nothing to Ψ_{pk}. Fig. 1(b) depicts the variation of Ψ_{pk} in the spectral domain at depth $d = 0.05$, 0.01, 0.15, and 0.20. In the spectral domain, Ψ_{pk} also exhibits a localization characteristic. As the depth d increases, Ψ_{pk} gradually shifts toward lower frequencies. Shallowly buried Ψ_{pk} has a broader spectral range and is more sensitive to high-frequency signals, whereas deeply buried Ψ_{pk} has a narrower spectral range and is more sensitive to low-frequency signals.

B. RBF Method

For the regional marine gravity field, assuming the effect of sea surface topography is neglected, the disturbing potential T can be expressed as

$$T = T_{\mathrm{GGM}} + T_{\mathrm{res}} \tag{3}$$

where T_{GGM} represents the medium-to-long wavelength signals expressed by the global gravity field model, and T_{res} denotes the residual short-wavelength signals. The residual disturbing potential T_{res} can be expanded as a series expression of Ψ_{pk}, as given by

$$T_{\mathrm{res}}(\boldsymbol{z}) = \sum_{i=1}^{K}\beta_i\boldsymbol{\Psi}_{\mathrm{pk}}(\boldsymbol{z},\boldsymbol{y}_i) \tag{4}$$

where K represents the number of Poisson kernel RBFs, β_i denotes the coefficient of the ith Ψ_{pk}, and $\boldsymbol{y}_i$ is the position vector of the ith Ψ_{pk}.

For convenience, define $\Delta g^1_{\mathrm{res}}$ as the residual altimetry gravity anomaly, and $\Delta g^2_{\mathrm{res}}$ as the residual shipborne gravity anomaly. The relationship between the two types of gravity observations and Ψ_{pk} is given as follows:

$$\Delta g^1_{\mathrm{res}}(\boldsymbol{z}_1) = \sum_{i=1}^{K}\beta_i\left(-\frac{2}{|\boldsymbol{z}_1|}\boldsymbol{\Psi}_{\mathrm{pk}}(\boldsymbol{z}_1,\boldsymbol{y}_i) - \frac{\partial\boldsymbol{\Psi}_{\mathrm{pk}}(\boldsymbol{z}_1,\boldsymbol{y}_i)}{\partial|\boldsymbol{z}_1|}\right) \tag{5}$$

$$\Delta g^2_{\mathrm{res}}(\boldsymbol{z}_2) = \sum_{i=1}^{K}\beta_i\left(-\frac{2}{|\boldsymbol{z}_2|}\boldsymbol{\Psi}_{\mathrm{pk}}(\boldsymbol{z}_2,\boldsymbol{y}_i) - \frac{\partial\boldsymbol{\Psi}_{\mathrm{pk}}(\boldsymbol{z}_2,\boldsymbol{y}_i)}{\partial|\boldsymbol{z}_2|}\right) \tag{6}$$

where $\boldsymbol{z_1}$ is the position vector of altimetry data, and $\boldsymbol{z_2}$ is the position vector of shipborne data.

By rearranging (5) and (6), the observation equation for the pth type of gravity observations can be expressed as

$$\boldsymbol{l}_p + \boldsymbol{e}_p = L_p T_{\mathrm{res}}(\boldsymbol{z_p}) = \sum_{i=1}^{K}\beta_i L_p\Psi_{\mathrm{pk}}(\boldsymbol{z_p},\boldsymbol{y}_i), \quad p = 1,2 \tag{7}$$

where $\boldsymbol{l}_p$ represents the pth type of gravity observations, $\boldsymbol{e}_p$ denotes the error vector for this type of observations, and L_p is the functional relationship between the gravity field parameters of this observation type and the disturbing potential, which can be derived from (5) and (6).

For the pth type of gravity observations, the corresponding error equation is given by

$$\boldsymbol{V}_p = \boldsymbol{A}_p\ \boldsymbol{X} - \boldsymbol{l}_p \tag{8}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

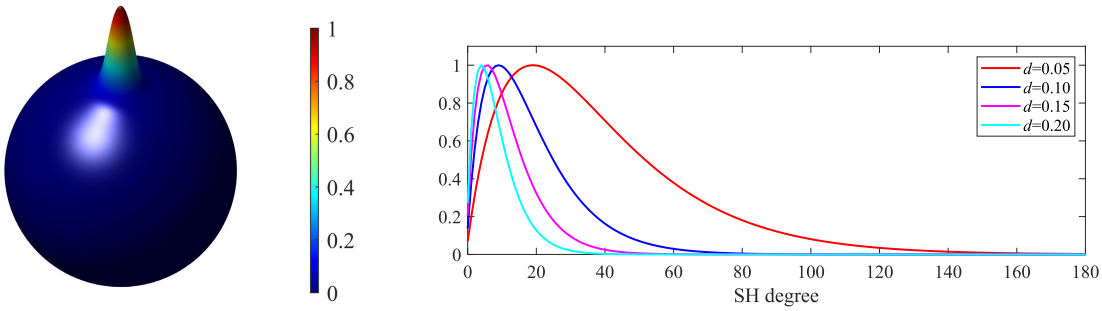


Fig. 1. Ψ_{pk} function. (a) Functional values of Ψpk at $d = 0.10$. (b) Spectral domain characteristics of Ψpk.

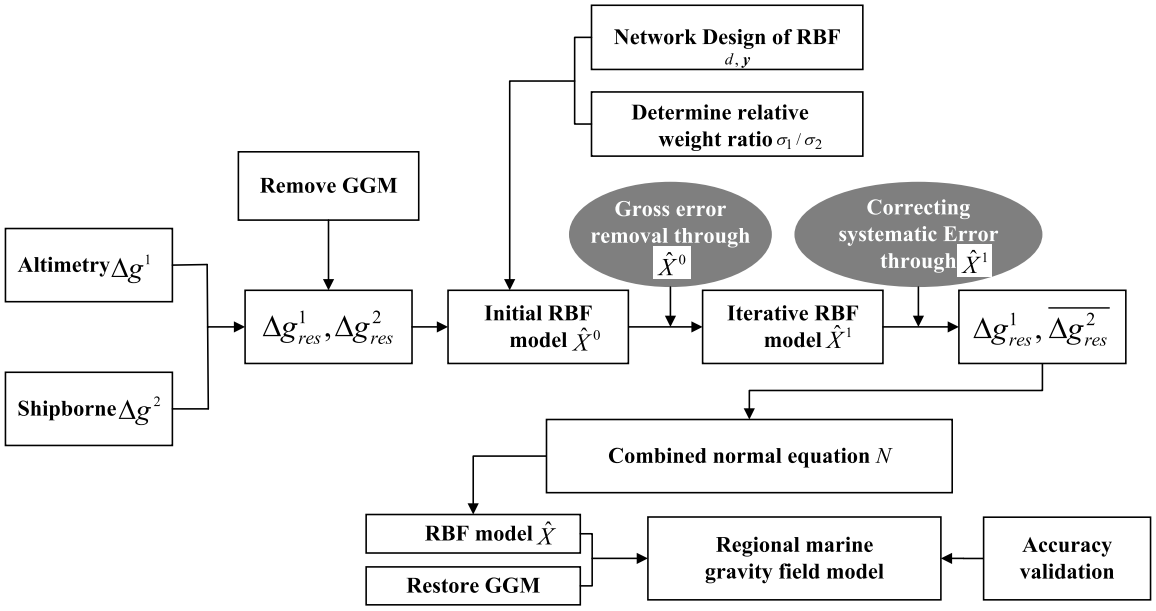


Fig. 2. Iterative RBF framework.

where $\boldsymbol{V}_p$ is the $t_p \times 1$ random error vector, $\boldsymbol{A}_p$ is the $t_p \times K$ design matrix, $\boldsymbol{X}$ is the $K \times 1$ parameter vector, $\boldsymbol{l}_p$ is the $t_p \times 1$ observations vector, and t_p denotes the number of observations. Consequently, the error equation for the fusion of various types of observations can be formulated as

$$\boldsymbol{V} = \boldsymbol{AX} - \boldsymbol{l}. \tag{9}$$

When the two types of gravity observations are uncorrelated, the error variance-covariance matrix is expressed as

$$\boldsymbol{C} = \begin{Bmatrix} \boldsymbol{C}_1 & 0 \\ 0 & \boldsymbol{C}_2 \end{Bmatrix} \tag{10}$$

where $E\{\boldsymbol{V}_p\} = 0$, $D\{\boldsymbol{V}_p\} = \boldsymbol{C}_p = \sigma_p^2 \boldsymbol{Q}_p$, $E\{\cdot\}$ and $D\{\cdot\}$ are the expectation and dispersion operators, $\boldsymbol{C}_p$ is the error variance-covariance matrix of the pth type of observations, σ_p^2 is the variance factor, $\boldsymbol{Q}_p$ is the cofactor matrix, and $\boldsymbol{Q}_p = \boldsymbol{I}$, where $\boldsymbol{I}$ is the identity matrix.

In practical modeling and calculations, challenges such as uneven distribution of multisource gravity data, data gaps, and different error characteristics may render the normal equations ill-conditioned. Therefore, it is necessary to apply an appropriate regularization method. Here, the Tikhonov regularization method [27], [33] is employed. Based on the least squares principle, for a given regularization parameter α, the regularized function model is expressed as

$$\Phi(\boldsymbol{X}) = \|\boldsymbol{V}\|^2_{\boldsymbol{C}^{-1}} + \alpha \|\boldsymbol{X}\|^2_{\boldsymbol{E}} \tag{11}$$

where $\|\boldsymbol{V}\|^2_{\boldsymbol{C}^{-1}} = \sum_{p=1}^{2} \|\boldsymbol{V}\|^2_{\boldsymbol{C}_p^{-1}}$, $\boldsymbol{E}$ is the regularization matrix and $\boldsymbol{E} = \boldsymbol{I}$.

By minimizing the regularized function model, the least-squares solution of the normal equation can be derived as [18]

$$\hat{\boldsymbol{X}} = \boldsymbol{N}^{-1}\boldsymbol{W} \tag{12}$$

where $\boldsymbol{N} = \sum_{p=1}^{2} \frac{1}{\sigma_p^2}\boldsymbol{N}_p + \alpha \boldsymbol{E}, \boldsymbol{N}_p = \boldsymbol{A}_p^T \boldsymbol{Q}_p^{-1} \boldsymbol{A}_p, \boldsymbol{W} = \sum_{p=1}^{2} \frac{1}{\sigma_p^2}\boldsymbol{W}_p, \boldsymbol{W}_p = \boldsymbol{A}_p^T \boldsymbol{Q}_p^{-1} \boldsymbol{l}_p$.

C. Iterative RBF Framework

The iterative RBF framework integrates the data quality control process with the gravity field modeling procedure to handle complex errors in multisource heterogeneous data and achieve the construction of high-precision regional gravity field models. This method involves repeatedly constructing RBF models and utilizing each iteratively updated model as a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

dynamic reference to detect gross errors and systematic errors in the data, thereby achieving "self-cleaning" of the data and "self-calibration" of the model. The shipborne data involved in this study were sourced from multiple surveying institutions, with some datasets being outdated and containing significant gross errors and systematic biases. Direct use of such data for modeling would lead to distortion in the final gravity field model. Therefore, employing the aforementioned method for preprocessing the shipborne data is a critical step to ensure model reliability. Fig. 2 illustrates the complete modeling workflow of this method, which mainly consists of two steps: data preprocessing and final model construction. The first step involves the design of the basis function network, the determination of relative weight ratios, and the iterative processing of shipborne data, where $\Delta\bar{g}^2_{\text{res}}$ represents the residual gravity anomaly of the shipborne data corrected by the RBF model. The second step involves constructing the combined normal equation and solving for the model coefficients. The specific workflow of this method is as follows.

The first phase aims to eliminate gross errors and systematic errors in the shipborne data through an iterative strategy. First, an initial RBF model is constructed by fusing altimetry and raw shipborne data $\boldsymbol{l}_2^0$. The model coefficients $\hat{X}^0$ are solved, and the initial residuals of all shipborne points are calculated as follows:

$$\boldsymbol{V}_2^0 = \boldsymbol{A}_2^0\hat{\boldsymbol{X}}^0 - \boldsymbol{l}_2^0 \tag{13}$$

where $\boldsymbol{V}_2^0$ is the initial residual vector; $\boldsymbol{A}_2^0$ is the design matrix of the raw shipborne data; $\boldsymbol{l}_2^0$ is the raw shipborne data.

Due to the influence of gross errors, the residuals $\boldsymbol{V}_2^0$ of the initial RBF model mainly reflects a combination of model errors and gross errors. Accordingly, a threshold is set at three times the standard deviation (3σ) of the model residuals to identify and eliminate observations that exceed this threshold. The RBF model is then updated using the subset of data with gross errors removed, and the steps of residual calculation and gross error elimination are repeated until no new gross errors are detected. Through this iterative process, the reference baseline for gross error identification is continuously refined, enabling more effective detection of gross errors that were initially masked. After the elimination of gross errors, the cleaned data subset $\boldsymbol{l}_2^1$ is used to update the RBF model once again, yielding the model coefficients $\hat{X}^1$. The residuals are then calculated as follows:

$$\boldsymbol{V}_2^1 = \boldsymbol{A}_2^1\hat{\boldsymbol{X}}^1 - \boldsymbol{l}_2^1 \tag{14}$$

where $\boldsymbol{V}_2^1$ represents the residual vector of the current RBF model.

Since significant gross errors have been removed, the residual systematic signals in $\boldsymbol{V}_2^1$ become apparent. By analyzing the spatial distribution characteristics of the residuals $\boldsymbol{V}_2^1$, the shipborne survey line$\boldsymbol{l}_2^s$ with systematic error and the corresponding model residuals $\boldsymbol{V}_2^s$ are identified. The systematic error Δ of this survey line is then estimated and corrected as follows:

$$\bar{\boldsymbol{l}}_2^s = \boldsymbol{l}_2^s + \boldsymbol{\Delta} \tag{15}$$

where $\bar{l}_2^s$ represents the corrected survey line, and Δ is taken as the mean of $\boldsymbol{V}_2^s$.

Based on (13)–(15), the first step of the iterative RBF framework is completed, yielding the processed shipborne data $\bar{l}_2$. The second phase involves constructing the final RBF model by fusing the preprocessed shipborne data $\bar{l}_2$ with satellite altimetry data $\boldsymbol{l}_1$. The observation equations for both datasets are established according to (5) and (6), respectively. A combined normal equation is constructed, and the final RBF model coefficients are determined using the least-squares method, ultimately yielding the regional marine gravity field model. Finally, the accuracy of the model is evaluated using external shipborne data that were not involved in the modeling process.

III. Study Region and Data

A. Study Region

The study regions are the Ogasawara Islands and the central-eastern Philippine Sea, both located in the northwest Pacific. The Ogasawara Islands region is located in (24°N–29.5°N, 139.5°E–146.5°E), as shown in Fig. 3(a). This region lies in the subduction zone between the Pacific Plate and the Philippine Sea Plate, characterized by frequent geological activities and complex seafloor topography. The western part of the region is dominated by a north-south trending chain of seamounts, with the southernmost point being the well-known Sulfur Islands. The central part is located along the western Pacific subduction zone, where the Izu-Ogasawara Ridge and Trench exist. The Izu-Ogasawara Trench connects to the Japan Trench to the north and the Mariana Trench to the south, with maximum depth reaching ~10 km, making it one of the major trenches in the western Pacific. The northeastern portion primarily consists of deep-sea plains, interspersed with east-west trending seamounts. The southeastern area is dominated by the Ogasawara Plateau and associated guyots. The central-eastern Philippine Sea region is located in (15°N–23°N, 132.5°E–140°E), as shown in Fig. 3(b). This region lies east of Luzon Island and south of the Ryukyu Islands, within the Philippine Sea Plate. It encompasses the West Philippine Sea Basin, the Kyushu-Palau Ridge, and the Parece Vela Basin from west to east, with the Oki-Daito Ridge marking the northwest boundary. The topography in this region is generally smooth, with numerous small geological features found in the eastern part. The Ogasawara Islands region exhibits complex topography and significant gravity anomaly variations, while the central-eastern Philippine Sea region is characterized by relatively simpler topography and smoother gravity anomaly variations. High-frequency signals containing finer details are only present in the central-eastern Kyushu-Palau Ridge and the Parece Vela Basin. Moreover, extensive marine surveys conducted by countries such as Japan and the United States in the northwest Pacific have generated a wealth of shipborne data, enriching the geological and geophysical understanding of this region.

B. Data

This study aims to refine the regional marine gravity field by fusing satellite altimetry and shipborne data. The data utilized includes: the global gravity field model, satellite altimetry-derived gravity anomalies, and shipborne gravity data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

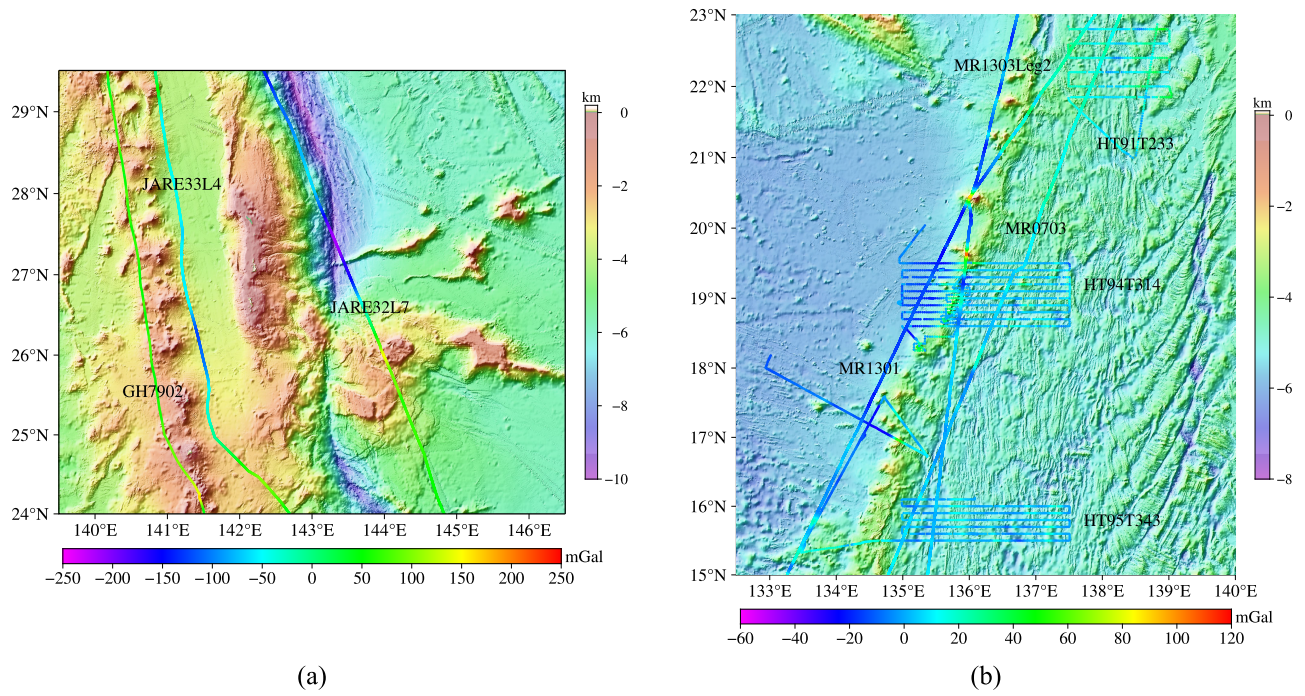


Fig. 3. Seafloor topography of the study regions (nine shipborne survey lines, such as GH7902, JARE33L4, and JARE32L7, were used for spectral analysis; topographic data sourced from SRTM15 +). (a) Ogasawara Islands. (b) Central-Eastern Philippine Sea.

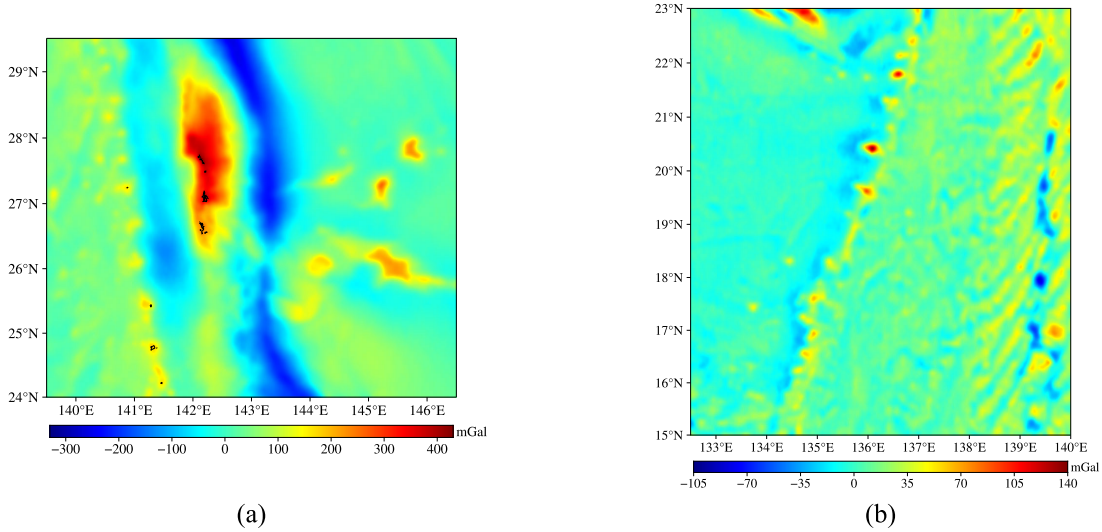
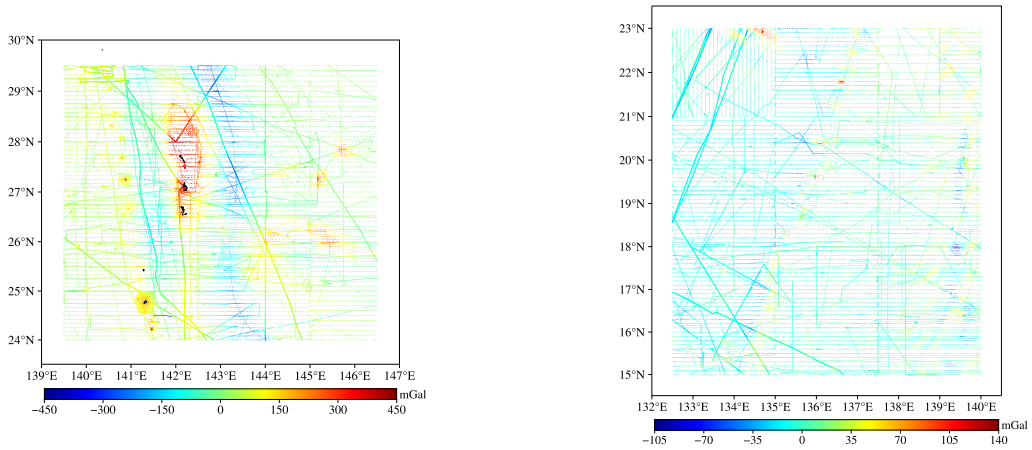


Fig. 4. SS 32.1 altimetry gravity anomaly. (a) Ogasawara Islands. (b) Central-Eastern Philippine Sea.

1) Global Gravity Field Model: Regional gravity field modeling usually requires the removal of medium-to-long wavelength signals. To achieve this, a global gravity field model is commonly employed as a reference to isolate short-wavelength signals. The GOSG02S [34] is a pure GOCE satellite gravity field model with a maximum degree of 300. It combines high-accuracy, full-cycle gravity gradient component observations from the GOCE satellite and satellite tracking data [35]. This model is publicly available through the International Centre for Global Earth Models (ICGEMs). We believe that the 200th degree GOSG02S model contains medium-to-long wavelength signals, which are then removed from both the altimetry and shipborne data.

2) Satellite Altimetry: The DTU17 (Technical University of Denmark) [36] and SS 32.1 (Sandwell and Smith) are global marine gravity field models derived from multiple generations of altimetry satellites. Developed by the Andersen group at DTU, DTU17 is available at $1' \times 1'$ and $2' \times 2'$ resolutions. It utilizes data from altimetry satellites such as T/P, ERS-2, Jason-1, SARAL/AltiKa, and CryoSat-2, with main improvements in accuracy for coastal and polar regions, as well as for short-wavelength gravity signals (10–15 km). The SS 32.1, released by the Sandwell group at the Scripps Institution of Oceanography (SIO), has a resolution of $1' \times 1'$. The SS 32.1 incorporates observations from altimetry satellites, including SARAL, CryoSat-2, Jason-1/2, Sentinel-3A/B, and ENVISAT.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 5. Shipborne gravity anomaly. (a) Ogasawara Islands. (b) Central-Eastern Philippine Sea.

TABLE I

STATISTICS OF CROSSOVER DISCREPANCIES IN SHIPBORNE DATA BEFORE AND AFTER PROCESSING WITH THE ITERATIVE RBF FRAMEWORK (UNIT: MGAL)

Study Region	Before / After Processing	Points	Min	Max	Mean	STD	M
Ogasawara Islands	Before Processing	6423	-618.339	982.223	-1.670	40.463	28.634
	After Processing	5715	-107.655	110.285	0.110	13.295	9.400
Central-eastern Philippine Sea	Before Processing	3963	-784.515	113.036	0.578	17.284	12.227
	After Processing	3685	-40.631	35.123	0.336	6.036	4.274

Compared to the DTU17 model, the SS 32.1 model uses longer periods of observations from satellites such as CryoSat-2, Jason-1/GM, and SARAL/AltiKa, adding observations from Jason-2, Sentinel-3A/B, and retracking waveforms for Jason-1/2, ENVISAT, AltiKa, and CryoSat in low-resolution mode (LRM), reducing the land-sea boundary effect. For this study, the $1' \times 1'$ resolution SS 32.1 altimetry gravity anomaly is selected as a modeling dataset. This dataset contains 138 930 points for the Ogasawara Islands region and 216 480 points for the central-eastern Philippine Sea region, as shown in Fig. 4.

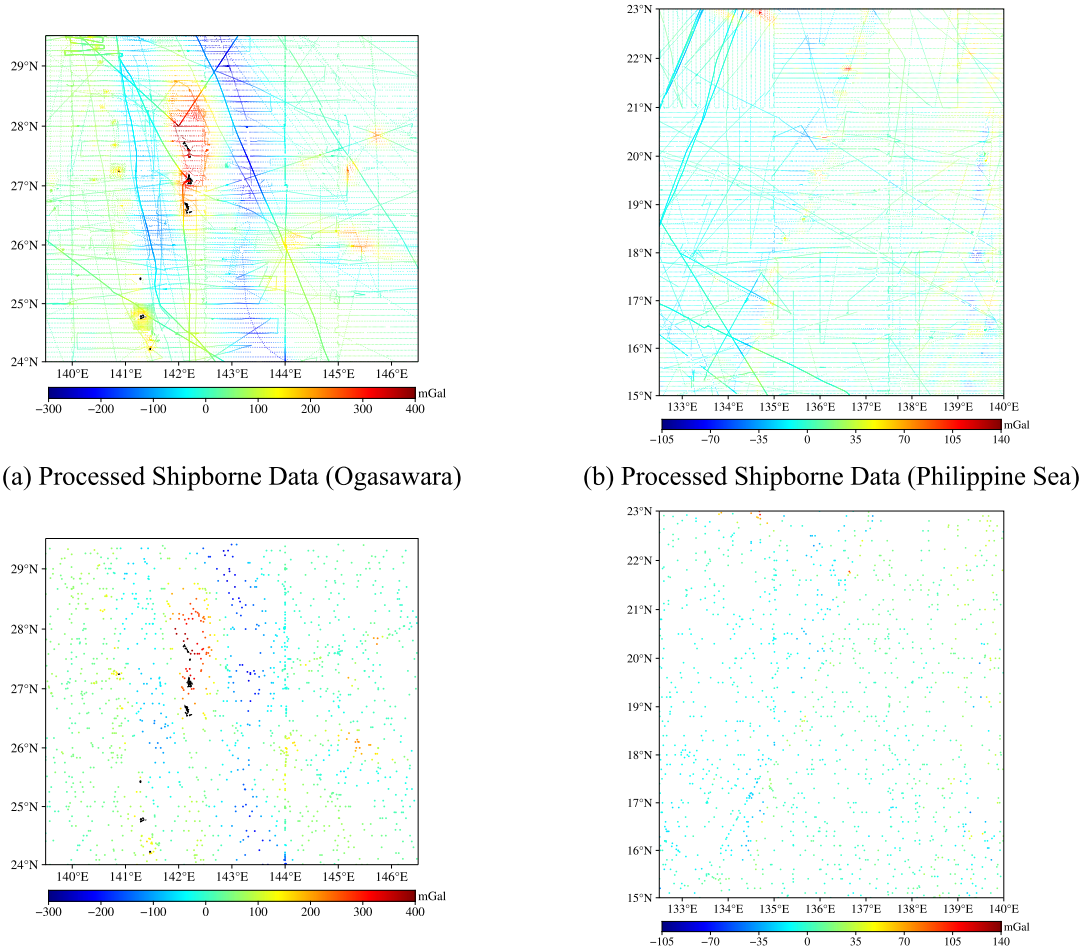
3) Shipborne: The shipborne data utilized in this study is sourced from the U.S. National Center for Environmental Information (NCEI), involving multiple surveying institutions, including Japan's Maritime Safety Agency (MSA), the University of Tokyo (UTokyo), the National Oceanic and Atmospheric Administration (NOAA), and the Hawaii Institute of Geophysics (HIG) at the University of Hawaii, among others. The dataset spans from the mid-1960s to the early 21st century, covering a period of over 30 years. The shipborne data have been subjected to Eötvös and drift corrections. In the Ogasawara Islands area, there are 56 shipborne survey lines comprising 53 403 points, as shown in Fig. 5(a). In the central-eastern Philippine Sea region, 73 shipborne survey lines are presented, containing 45 459 points, as shown in Fig. 5(b).

IV. RESULTS

The RBF method for modeling requires the selection of an appropriate basis function network, which significantly influences the stability and accuracy of the model's solution. RBF networks incorporate both planar positions and depth parameters. Planar positions are typically selected from a grid model such as the Reuter grid [37], [38], Driscoll-Heal grid, and Fibonacci grid [8]. The Reuter grid features equidistant distribution between adjacent points and is chosen to define the planar positions of the basis functions. Depth parameters are generally determined using the generalized cross-validation (GCV) method [27]. However, the GCV approach can pose challenges, such as difficulties in identifying extreme points and relatively low computational efficiency. Therefore, we configured various combinations of the number of basis functions and depth, using the standard deviation of model residuals as the selection criterion. The combination with the smallest standard deviation was chosen as the optimal basis function network. In addition, the variance component estimation (VCE) method is commonly employed to determine the relative weights of multisource data. However, this method, which estimates unit weight variance based on residuals, is sensitive to factors such as resolution [39], quantity [18], and spatial distribution. Under complex data conditions, it may not provide reasonable weightings. Therefore, the optimal relative weight ratio σ_1/σ_2 for altimetry and shipborne data was determined through multiple modeling tests. Here, σ_1 represents altimetry, and σ_2 represents shipborne. We then determined the appropriate basis function network and relative weight ratio. For the Ogasawara Islands region, the basis function network has a resolution of 4 km (26 604 RBFs), a depth of 20 km, and $\sigma_1/\sigma_2 = 2.5$. For the central-eastern Philippine Sea region, the resolution is 5 km (28 082 RBFs), the depth is 20 km, and $\sigma_1/\sigma_2 = 12.0$.

After determining the appropriate basis function network and optimal relative weight ratio, we completed the first step of the iterative RBF framework and obtained the shipborne data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.



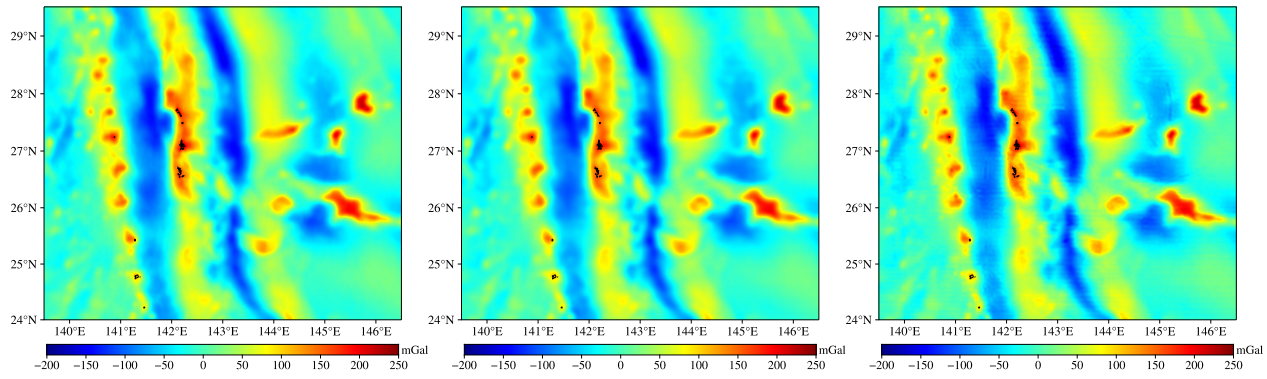
(a) Processed Shipborne Data (Ogasawara)

(b) Processed Shipborne Data (Philippine Sea)

(c)Validation Data (Ogasawara)

(d) Validation Data (Philippine Sea)

Fig. 6. Shipborne data processed by the iterative RBF framework and the validation datasets. (a) All shipborne data processed by the iterative RBF framework in the Ogasawara Islands region. (b) All shipborne data processed by the iterative RBF framework in the central-eastern Philippines region. (c) Independent shipborne data for external validation in the Ogasawara Islands region. (d) Independent shipborne data for external validation in the central-eastern Philippines region.



(a) (b) (c)

Fig. 7. Gravity anomaly model values in the Ogasawara Islands. (a) SS 32.1 model. (b) LSC model. (c) RBF model.

processed by the RBF model (see Fig. 6). In the Ogasawara Islands region, 51 137 shipborne data points were acquired [see Fig. 6(a)], with 2266 data points eliminated, resulting in a rejection rate of 4.24%. In the central-eastern Philippine Sea region, 44 030 shipborne data points were obtained [see Fig. 6(b)], with 1429 data points eliminated, corresponding to a rejection rate of 3.14%. To analyze the changes in the quality of shipborne data before and after processing with the iterative RBF framework, crossover discrepancies of the shipborne data were calculated and statistically analyzed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

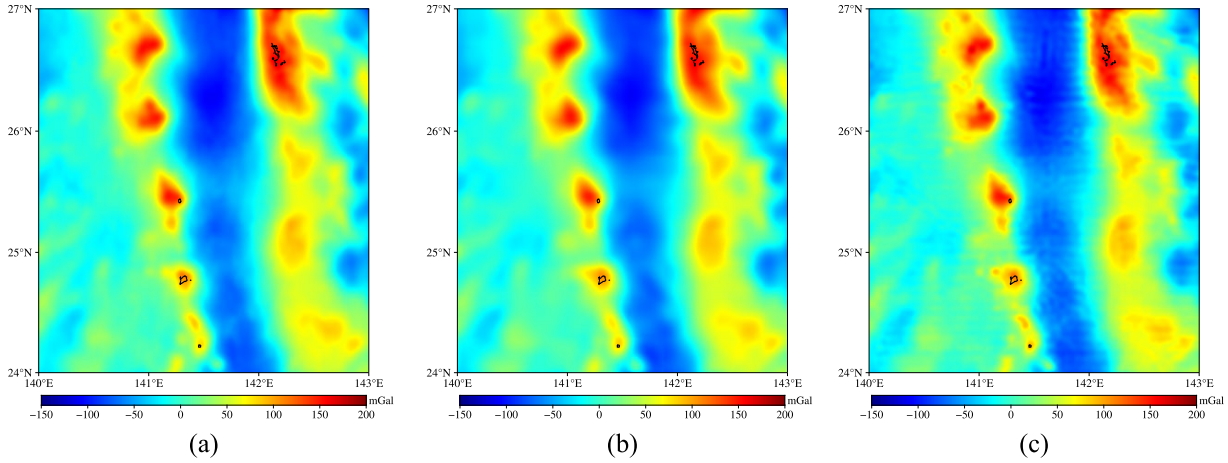


Fig. 8. Gravity anomaly model values in the Izu-Bonin arc and adjacent seas. (a) SS 32.1 model. (b) LSC model. (c) RBF model.

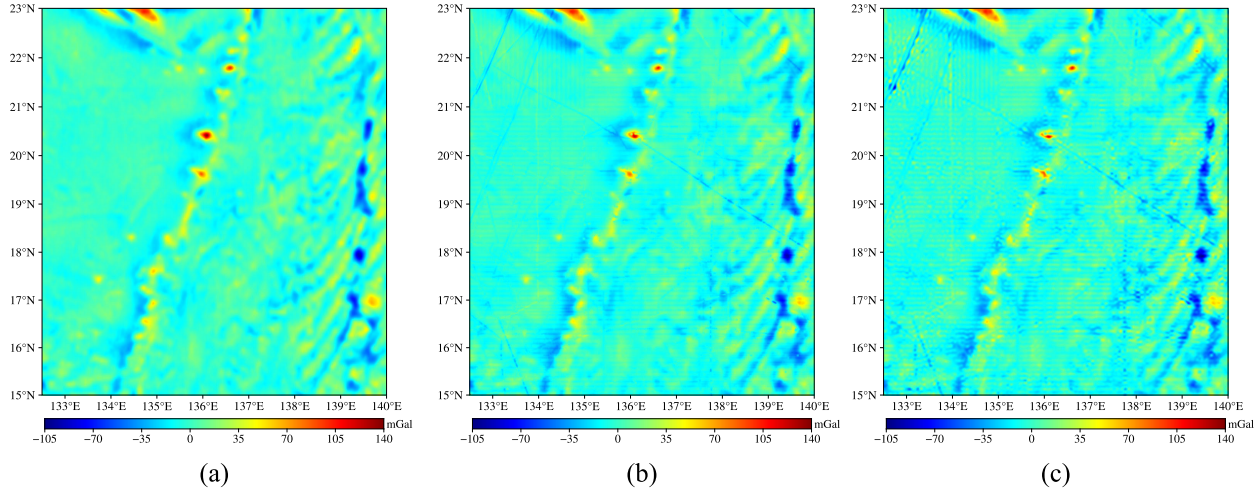


Fig. 9. Gravity anomaly model values in the central-eastern Philippine Sea. (a) SS 32.1 model. (b) LSC model. (c) RBF model.

before and after processing. The formula for estimating the internal consistency accuracy of shipborne data using the gravity anomaly discrepancy at the crossover points of survey lines i and j is as follows:

$$M_{\Delta g}^{inter} = \pm \sqrt{\frac{\sum_{k=1}^{N} \left(\Delta g_i - \Delta g_j\right)_k^2}{2N}} \tag{16}$$

where $M_{\Delta g}^{inter}$ represents the internal consistency accuracy of observation values; Δg_i and Δg_j denote the gravity anomalies at the crossover point from survey lines i and j, respectively, and N is the number of crossover points between the survey lines. The statistical results of crossover discrepancies for shipborne data before and after processing with the iterative RBF framework are shown in Table I. In the Ogasawara Islands region, a total of 6423 crossover points were identified in the raw shipborne data, while 5715 crossover points remained after processing. All statistical measures of the crossover discrepancies showed varying degrees of reduction after processing. In particular, the standard deviation of discrepancies decreased significantly from 40.463 to 13.295 mGal. The proportion of crossover discrepancies within the ±15 mGal range increased from 86.64% to 96.17%. The accuracy of the shipborne data improved from approximately 28.63 mGal to about 9.40 mGal, representing an accuracy enhancement of approximately 67.17%. In the central-eastern Philippine Sea region, 3963 crossover points were identified in the raw data, compared to 3685 after processing. Similar to the results in the Ogasawara Islands region, all statistical measures of crossover discrepancies demonstrated reductions after processing with the iterative RBF framework. Notably, the standard deviation of crossover discrepancies decreased markedly from 17.284 to 6.036 mGal. The proportion of discrepancies within the ±15-mGal range rose from 91.07% to 96.77%. The accuracy of the shipborne data improved from approximately 12.23 mGal to about 4.27 mGal, reflecting an accuracy improvement of approximately 65.04%. The crossover analysis results of the shipborne data indicate that the accuracy of the data in both sea regions was substantially enhanced after processing with the iterative RBF framework. Using the processed shipborne data, we completed the second step of the iterative RBF framework and constructed the regional marine gravity field fusion models. Before constructing the fusion models, a portion of the shipborne data was uniformly removed to serve as external validation data, ensuring that it did not contribute to the modeling process. The remaining shipborne data were utilized

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

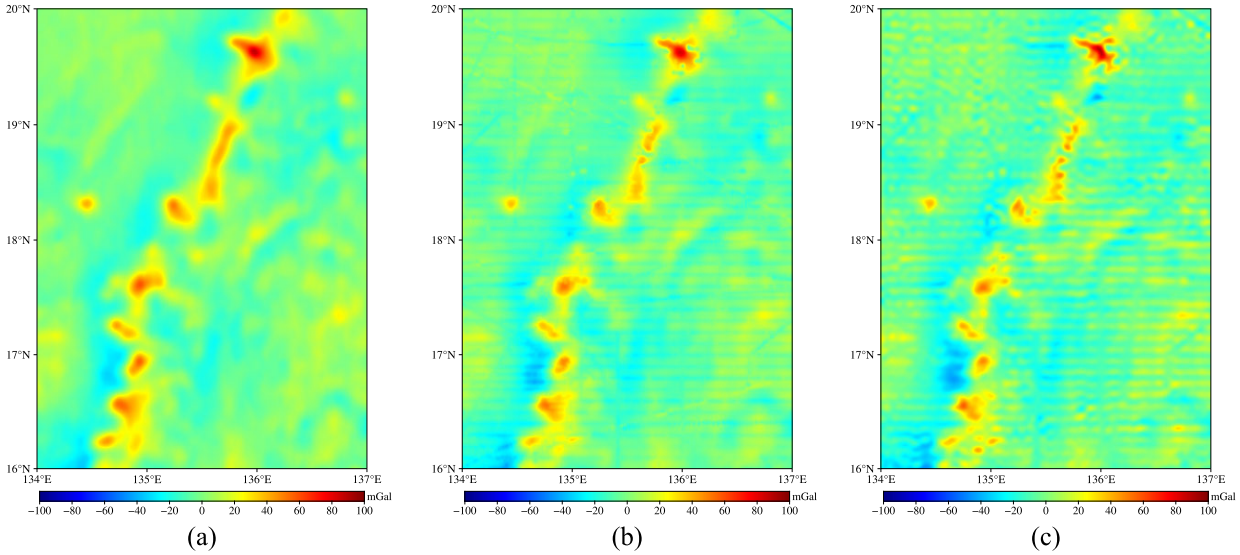


Fig. 10. Gravity anomaly model values in the Kyushu-Palau Ridge and adjacent seas. (a) SS 32.1 model. (b) LSC model. (c) RBF model.

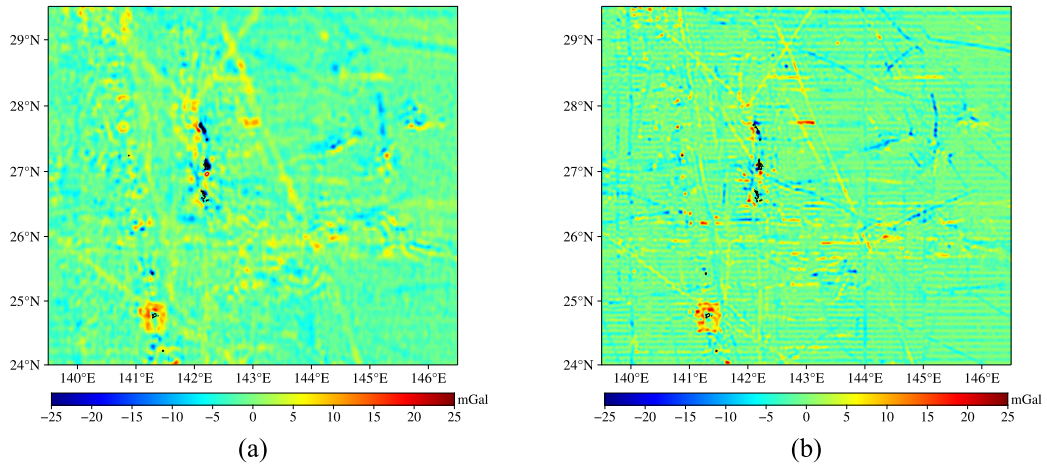


Fig. 11. Distribution of residuals for the fusion models in the Ogasawara Islands region. (a) LSC model. (b) RBF model.

to construct the fusion models. In the Ogasawara Islands region, a total of 1516 shipborne data points were excluded for use as external validation data [see Fig. 6(c)]. In the central-eastern Philippine Sea region, a total of 1350 shipborne data points were excluded for use as external validation data [see Fig. 6(d)]. Furthermore, the LSC method was also employed to fuse the altimetry and shipborne data, utilizing a local covariance model based on the logarithmic covariance function [40]. The relative weight ratio of the two data types remained consistent with the iterative RBF approach.

A. Regional Marine Gravity Field Fusion Model

1) Fusion Models: The gravity field fusion models for the Ogasawara Islands region are shown in Fig. 7. In comparison to the SS 32.1 residual altimetry gravity anomaly shown in Fig. 7(a), the fusion models depicted in Fig. 7(b) and (c) effectively recover the residual altimetry signal. The difference between the LSC model shown in Fig. 7(b) and the SS 32.1 model is relatively small, whereas the RBF model shown in Fig. 7(c) exhibits more pronounced features in high-frequency signal regions. The RBF model reveals more detailed local gravity field signals, particularly along the Izu-Bonin arc in the western part of the region and the Izu-Ogasawara Ridge in the central part. We present the values of each model in the Izu-Bonin arc and adjacent sea areas, as shown in Fig. 8. Compared to the SS32.1 and LSC models shown in Fig. 8(a) and (b), the RBF model in Fig. 8(c) exhibits significantly enhanced local anomaly signals generated by seamounts, islands, and ridges. The detailed signals are richer, allowing for clearer identification of smaller mass sources such as seamounts and islands. Furthermore, the contour lines of the gravity anomaly regions are more distinct. The fusion gravity field models for the central-eastern Philippine Sea region are shown in Fig. 9. Compared to the SS 32.1 in Fig. 9(a), the fusion models in Fig. 9(b) and (c) also reveal more detailed local gravity field signals, especially in the Kyushu-Palau Ridge, the Parece Vela Basin, and the Oki-Daito Ridge. We present the values of each model in the Kyushu-Palau Ridge and adjacent sea areas, as shown in Fig. 10. Compared to the SS32.1 models shown in Fig. 10(a), the LSC and RBF models shown in Fig. 10(b) and (c) demonstrate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

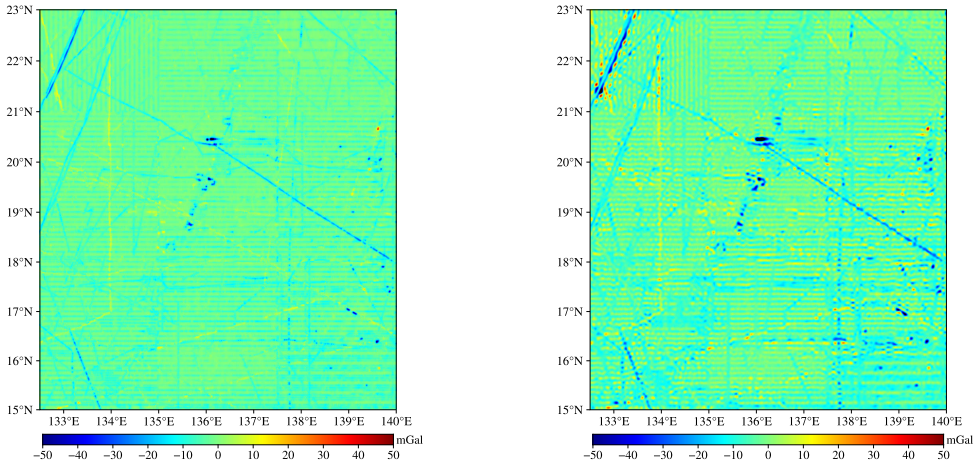


Fig. 12. Distribution of residuals for the fusion models in the central-eastern Philippine Sea region. (a) LSC model. (b) RBF model.

TABLE II

RESIDUAL STATISTICS OF REGIONAL MARINE GRAVITY FIELD FUSION MODELS (UNIT: MGAL)

Study Region	Type	Min	Max	Mean	STD	RMS
Ogasawara Islands	LSC model-Altimetry	-44.08	27.463	-1.162	2.563	2.815
	LSC model-Shipborne	-38.524	29.838	0.506	4.188	4.219
	RBF model-Altimetry	-32.320	24.189	-0.844	2.637	2.768
	RBF model-Shipborne	-16.728	17.940	0.378	2.707	2.733
Central-eastern Philippine Sea	LSC model-Altimetry	-96.825	39.132	-3.110	3.356	4.575
	LSC model-Shipborne	-4.932	6.051	0.110	0.365	0.381
	RBF model-Altimetry	-108.083	45.316	-3.751	5.674	6.801
	RBF model-Shipborne	-14.211	9.900	0.132	1.487	1.493

TABLE III

STATISTICAL ANALYSIS OF DIFFERENCES BETWEEN MARINE GRAVITY MODELS AND VALIDATION DATA (UNIT: MGAL)

Study Region	Model	Min	Max	Mean	STD	RMS
Ogasawara Islands	DTU17 model	-35.567	23.466	2.952	4.128	5.073
	SS 32.1 model	-28.527	23.000	3.047	3.538	4.668
	non-iterative RBF model	-43.079	66.268	1.553	6.297	6.484
	LSC model	-29.652	21.535	1.784	3.813	4.208
	iterative RBF model	-16.770	23.943	1.674	2.951	3.392
Central-eastern Philippine Sea	DTU17 model	-25.183	111.056	6.920	4.934	8.498
	SS 32.1 model	-27.832	113.544	7.066	4.961	8.633
	non-iterative RBF model	-23.063	35.061	0.629	3.544	3.598
	LSC model	-24.038	92.612	3.441	4.259	5.474
	iterative RBF model	-19.477	35.055	0.681	2.949	3.026

stronger localization of gravity anomaly signals, particularly in high-frequency signal regions such as seamounts. Along the Kyushu-Palau Ridge, both the LSC and RBF models exhibit richer, detailed features. Compared to the LSC model in Fig. 10(b), the RBF model in Fig. 10(c) demonstrates an enhanced capability to represent more localized and detailed gravity signals, enabling clear identification of anomaly signals generated by smaller seamounts. The fusion of altimetry and shipborne data in the regional marine context effectively compensates for the short-wavelength components missing in the SS 32.1 model.

2) Residual Analysis: The residuals between the regional marine gravity field fusion models and the SS 32.1 model were calculated for both the Ogasawara Islands and the central-eastern Philippine Sea regions, as shown in Figs. 11 and 12. Fig. 11 illustrates the distribution of residuals between the fusion models and the SS 32.1 model in the Ogasawara Islands region. The residuals are predominantly concentrated near the islands and along several shipborne survey lines. Notably, the Izu-Bonin arc in the western part of the region and the Ogasawara Islands in the central-western part exhibit larger residuals. These residuals are attributed primarily to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

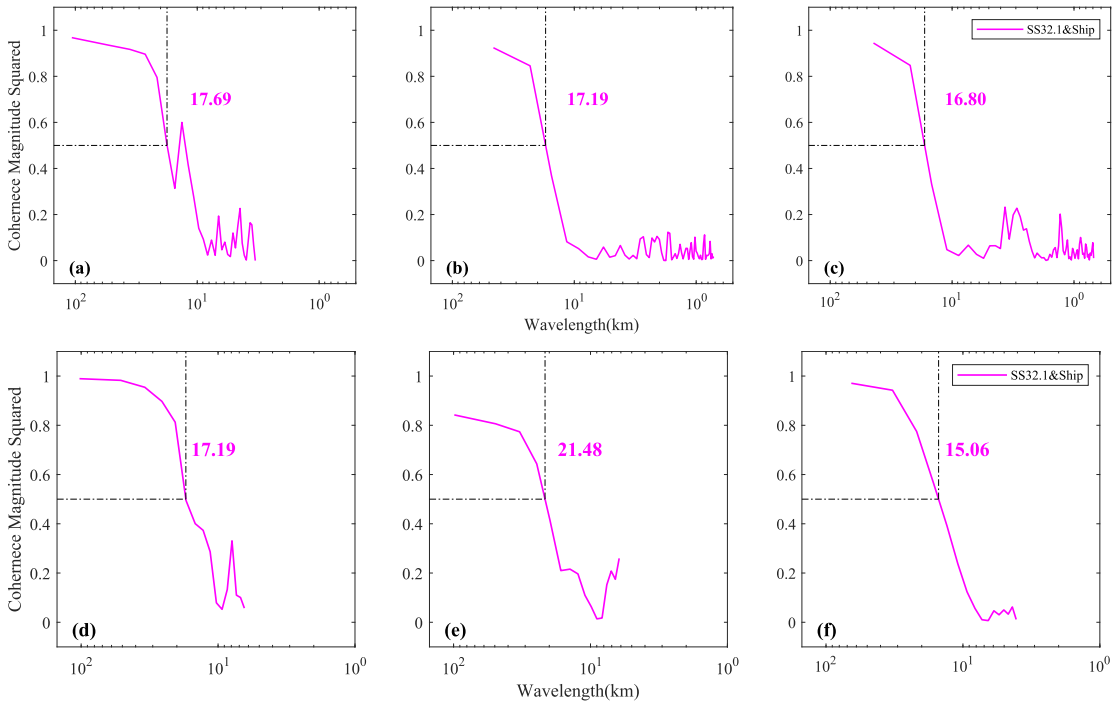


Fig. 13. Coherence between the SS 32.1 model and shipborne data. (a) GH7902. (b) JARE32L7. (c) JARE33L4. (d) HT91T233. (e) HT94T314. (f) HT95T343.

high-frequency signals from the shipborne data and errors in the fusion models. Compared to Fig. 11(a), the residuals in Fig. 11(b) show stronger localization, whereas the residuals in Fig. 11(a) are relatively smooth. Fig. 12 shows the residuals between the fusion models and the SS 32.1 model for the central-eastern Philippine Sea region. The residuals are chiefly distributed around the Kyushu-Palau Ridge, the Parece Vela Basin, and several shipborne survey lines. Similar to the Ogasawara Islands region, the residual signals here are predominantly caused by high-frequency components of the shipborne data and errors in the fusion models. Furthermore, as observed in Fig. 11(a) and (b), the residuals in Fig. 12(b) for the RBF model and SS 32.1 model also exhibit more localization, while those in Fig. 12(a) for the LSC and SS 32.1 models are smoother. In addition, the residuals between the fusion models and the shipborne data were computed, and a statistical analysis of all residuals was performed, as shown in Table II. In the Ogasawara Islands region, both LSC and RBF models showed smaller residuals with the SS 32.1 model, but relatively larger residuals with shipborne data. Notably, the RBF model demonstrated significant advantages in this region, with a standard deviation of residuals as low as 2.707 mGal compared to shipborne data, substantially lower than the 4.188 mGal of the LSC model. In the eastern Central Philippine Sea region, the residuals between LSC/RBF models and the SS 32.1 model increased, while residuals with shipborne data decreased, which is related to the quality of shipborne data and the relative weighting of the two data types. In this region, the LSC model exhibited superior performance, with a residual standard deviation of only 0.365 mGal compared to shipborne data, significantly lower than the 1.487 mGal of the RBF model. The Ogasawara Islands region features large-scale tectonic units and complex topographic characteristics, making it less suitable for the LSC method. In the east-central Philippine Sea, a region of plains and basins, the LSC model proved significantly advantageous for the shipborne data already incorporated in the modeling. Compared to the LSC method, the RBF method exhibited more stable performance, delivering satisfactory results in both regions with distinct topographic features.

B. Accuracy Assessment

We primarily analyzed the accuracy of the regional marine gravity field fusion models, including the LSC and RBF models. In addition, the DTU17 and SS 32.1 models were chosen for comparative analysis. By calculating the discrepancies between each model (DTU17, SS 32.1, LSC, and RBF) and the external validation data [as shown in Fig. 6(c) and (d)], statistical results were obtained (as shown in Table III). In the Ogasawara Islands region, the standard deviation of the noniterative RBF model is 6.297 mGal. After applying the iterative strategy, the standard deviation decreased by approximately 3.35 mGal, resulting in a significant accuracy improvement of about 53.14%. Among the four models, the RBF model achieves the highest accuracy, with a standard deviation of 2.951 mGal. In particular, the LSC model shows a 0.32 mGal reduction in standard deviation compared to DTU17. However, it exhibits no improvement in standard deviation against SS 32.1, even though the RMS value is 0.46 mGal lower. The RBF model demonstrates reductions in standard deviation of approximately 1.18 and 0.59 mGal compared to DTU17 and SS 32.1, respectively. Compared with the external validation data, the accuracy of the RBF model is improved by approximately 28.59% and 16.68%

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

over the DTU17 and SS 32.1 models, respectively. Validation results in the east-central Philippine Sea further confirmed this trend: the standard deviation of the noniterative RBF model was 3.544 mGal, which decreased by approximately 0.60 mGal after applying the iterative strategy, resulting in an accuracy improvement of about 16.79%. As in the Ogasawara Islands region, the RBF model again performed the best among the four models, with a standard deviation of 2.949 mGal. In addition, the LSC model reduces the standard deviation by 0.68 and 0.70 mGal compared to DTU17 and SS 32.1, respectively. The RBF model reduces the standard deviation by approximately 1.99 and 2.01 mGal compared to DTU17 and SS 32.1, respectively. In terms of accuracy improvement, the LSC model achieves approximately 13.68% and 14.15% enhancement compared to DTU17 and SS 32.1, while the RBF model shows improvements of approximately 40.23% and 40.56% compared to DTU17 and SS 32.1, respectively. The Ogasawara Islands and the east-central Philippine Sea exhibit distinctly different topographic characteristics, and there are disparities in the quality of shipborne data. Nevertheless, the RBF fusion model demonstrated the highest accuracy and superior stability in both regions. This result clearly demonstrates the advantages of the iterative RBF framework in handling multisource heterogeneous data and regional marine gravity field modeling.

C. Spectral Analysis

Spectral analysis offers a powerful method for studying the frequency domain characteristics of signals. The coherence function and power spectral density (PSD) provide essential insights into the degree of correlation between signals and the distribution of power across different frequencies. To analyze the spectral properties of the fusion models, we calculated the PSD along the track of the shipborne data for the DTU17, SS 32.1, LSC, and RBF models, employing the Welch method [41]. In addition, the coherence spectra between the shipborne data (serving as the reference) and the DTU17, SS 32.1, LSC, and RBF models were computed. This analysis enabled the determination of the effective wavelengths for each model. As shown in (17), the coherence function can assess the similarity between two signals and determine the effective wavelength. According to Marks' criteria for assessing the wavelength resolution of the geoid, the resolution standard is defined as the spatial wavelength at which the root mean square consistency drops to 0.5 [42], [43]. This criterion was also applied to assess the wavelength resolution of gravity anomalies

$$C_{xy}(\omega) = \frac{|P_{xy}(\omega)|^2}{P_{xx}(\omega)\,P_{yy}(\omega)} \tag{17}$$

where $P_{xy}(\omega)$ represents the cross-PSD of $x(t)$ and $y(t)$; $P_{xx}(\omega)$ and $P_{yy}(\omega)$ denote the auto-power spectral densities of $x(t)$ and $y(t)$, respectively.

Using discrete shipborne data to construct the LSC and RBF models, we selected specific survey lines (see Fig. 3) for the spectral analysis of these fusion models. The spectral analysis data shown in Fig. 3 were not corrected by the RBF model. In the Ogasawara Islands region, three survey lines were chosen. In Fig. 3(a), from west to east, these are the GH7902, JARE33L4, and JARE32L7 lines, which predominantly follow a north-south orientation, covering the entire Ogasawara Islands region. In the central-eastern Philippine Sea region, there are six shipborne survey lines, as shown in Fig. 3(b). These include HT91T233, HT94T314, and HT95T343, which are distributed along the Kyushu-Palau Ridge, spanning the northern, central, and southern parts of the region, respectively. These data contributed to the construction of the fusion models. In addition, the MR0703, MR1301, and MR1303Leg2 survey lines, published by Japan Agency for Marine Earth Science and Technology (JAMSTEC), provide extensive coverage from north to south [as shown in Fig. 3(b)]. These three lines were measured more recently and have higher accuracy. The data derived from these lines have been corrected for Eötvös and drift. They were not incorporated into the construction of the fusion models, thereby serving as independent external survey lines.

First, the coherence between the SS 32.1 model and shipborne data was calculated in both regions, and the cutoff wavelength of the SS 32.1 model was determined (as shown in Fig. 13). Then, the power spectra of the DTU17, SS 32.1, LSC, and RBF models, as well as the shipborne data, were calculated and compared (as shown in Fig. 14). In Fig. 14, the power spectra of the DTU17 and SS 32.1 models are nearly identical across the entire wavelength range. However, significant discrepancies are observed between these models and the shipborne data below the cutoff wavelength. In the Ogasawara Islands region, the LSC model in the high-frequency short-wavelength range (6–17 km) exhibits near consistency with the DTU17 model, and is slightly lower than the SS 32.1 model. For the JARE32L7 and JARE33L4 survey lines, the LSC model in the 2–6 km wavelength range is closer to the shipborne data and is not lower than the DTU17, SS 32.1, and RBF models. The RBF model effectively complements the altimetry signal in the short-wavelength range (approximately 6–17 km). In particular, across the 8–17 km wavelength range, the power of the RBF model closely matches that of the shipborne data. Between the 6 and 8 km wavelength range, although the RBF model shows some power attenuation relative to the shipborne data, its power remains superior to the DTU17, SS 32.1, and LSC models. In the shorter 2–6 km wavelength range, the RBF model remains generally consistent with the DTU17 and SS 32.1 models. In the central-eastern Philippine Sea region, the LSC model shows broad agreement with the shipborne data across the full wavelength range. In the 8–17 km wavelength range, both the LSC and RBF models exhibit nearly identical power spectra to the shipborne data. However, below 8 km, the LSC model is closer to the shipborne data. In the 6–8-km range, although the RBF model experiences a gradual power decrease, its signal strength remains higher than that of the DTU17 and SS 32.1 models.

The coherence between the DTU17, SS 32.1, LSC, and RBF models and shipborne data was calculated in both regions, as shown in Fig. 15. The coherence between the DTU17, SS 32.1 models, and the shipborne data is generally consistent across the entire wavelength range. In the Ogasawara Islands region, the LSC model exhibits slightly higher values than the DTU17 and SS 32.1 models within

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

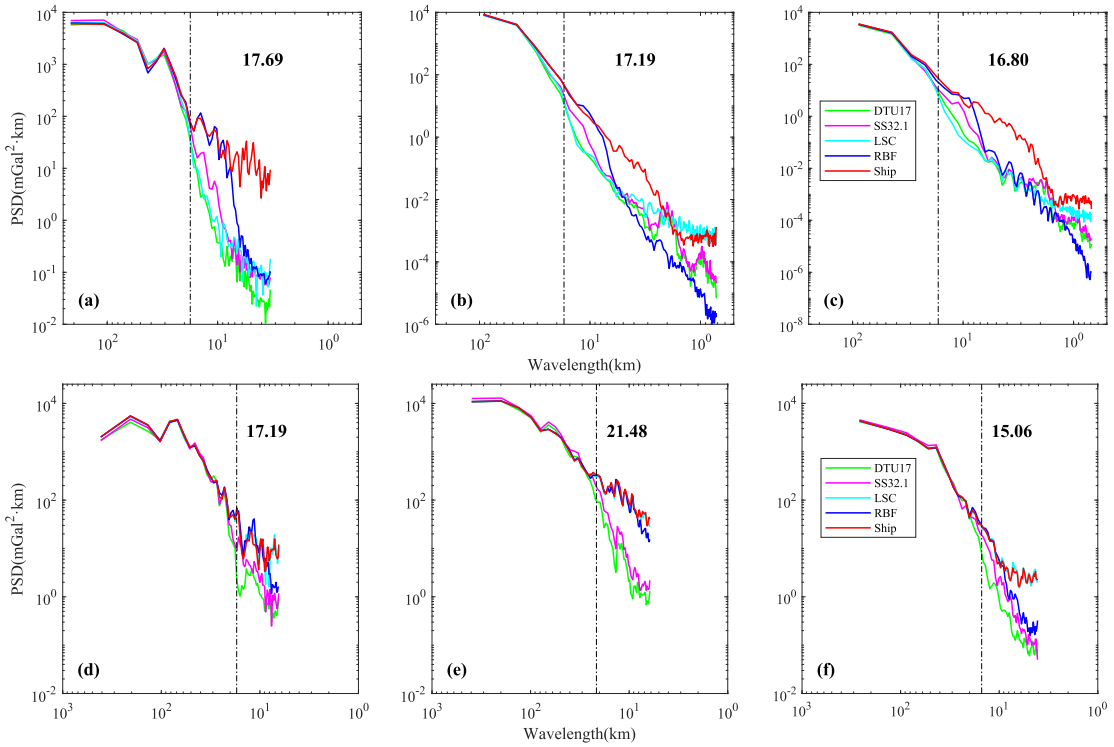


Fig. 14. Power spectral of various gravity data. (a) GH7902. (b) JARE32L7. (c) JARE33L4. (d) HT91T233. (e) HT94T314. (f) HT95T343.

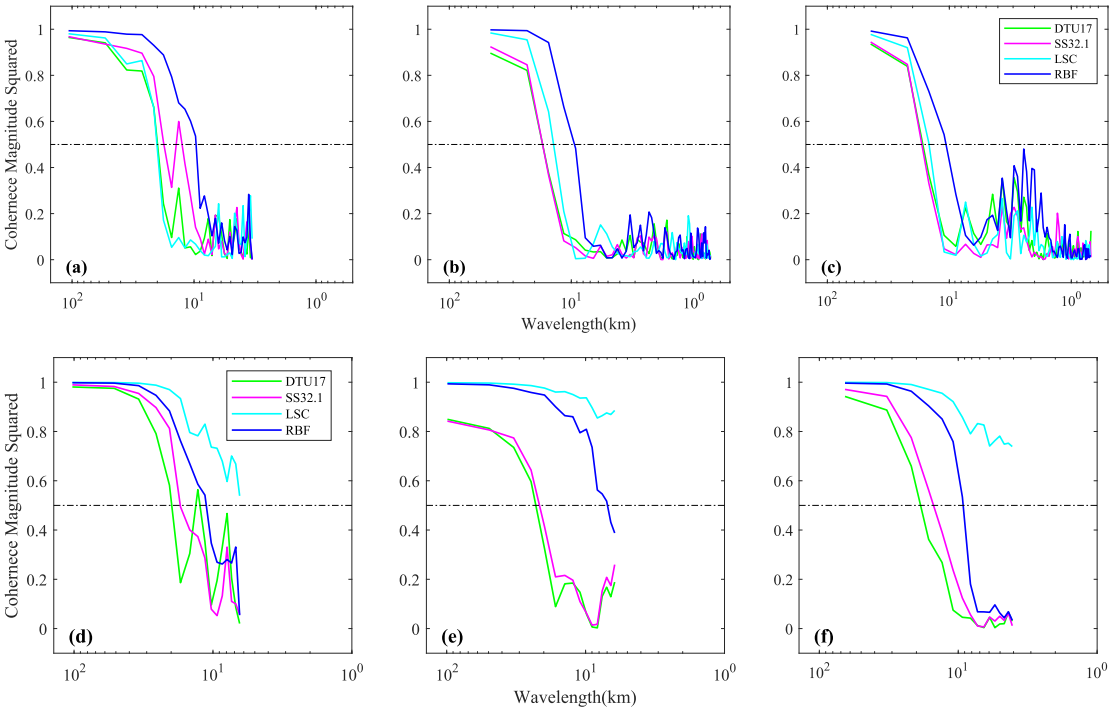


Fig. 15. Coherence between various marine gravity field models and shipborne data. (a) GH7902. (b) JARE32L7. (c) JARE33L4. (d) HT91T233. (e) HT94T314. (f) HT95T343.

the 12–17 km wavelength range. However, on the GH7902 survey line, the coherence is similar to that of the DTU17 model. Below approximately 12 km, the LSC model shows near-identical coherence to both the DTU17 and SS 32.1 models. In addition, the effective wavelength of the LSC model reduces to about 14.5 km, surpassing the DTU17 and SS 32.1 models. The RBF model demonstrates a significantly higher coherence than the DTU17, SS 32.1, and LSC models within the 8–17 km wavelength range, which is consistent with the power spectral analysis results. Below 8 km, the RBF model

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

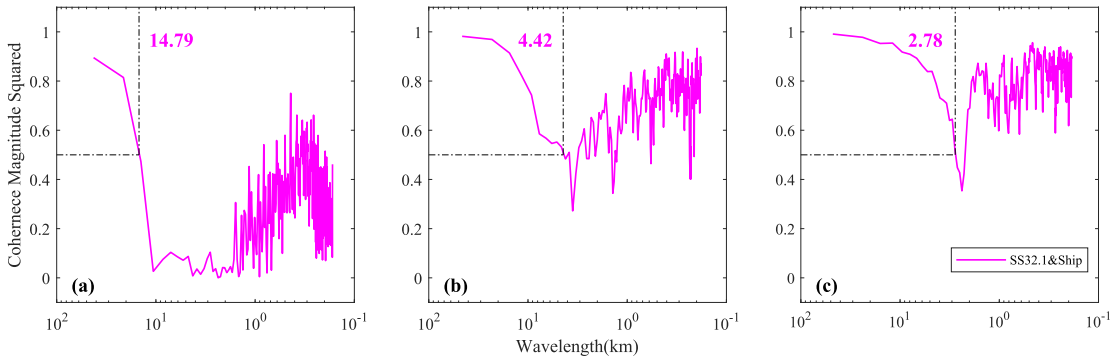


Fig. 16. Coherence between the SS 32.1 model and external survey lines. (a) MR0703. (b) MR1301. (c) MR1303.

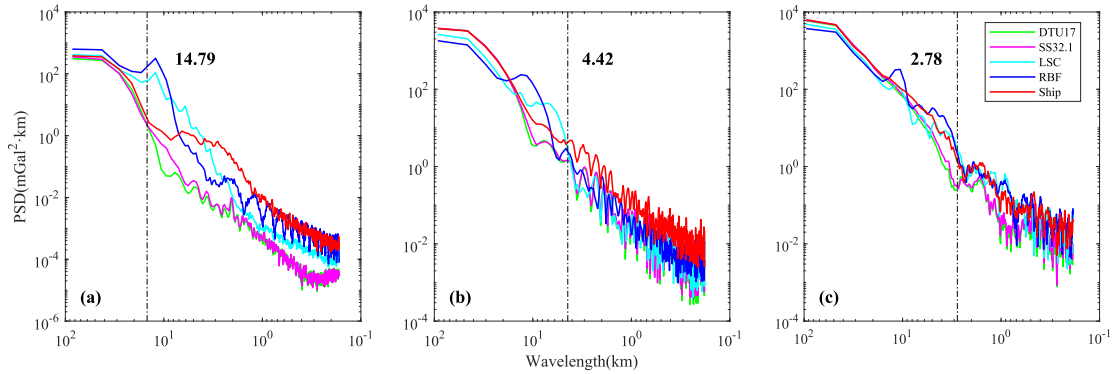


Fig. 17. Power spectral of various gravity data in the central-eastern Philippine Sea region (on the external survey lines). (a) MR0703. (b) MR1301. (c) MR1303.

TABLE IV
ACCURACY STATISTICS OF THE FUSION MODELS ON EXTERNAL SURVEY LINES (UNIT: MGAL)

Model	Survey line name	Min	Max	Mean	STD	RMS
LSC model	MR0703	-23.554	4.129	-2.738	3.074	4.116
	MR1301	-68.048	23.373	-3.180	4.871	5.817
	MR1303Leg2	-61.379	5.562	-3.016	3.880	4.914
RBF model	MR0703	-10.564	5.360	-1.889	2.832	3.404
	MR1301	-12.480	6.940	-2.375	3.034	3.852
	MR1303Leg2	-18.710	7.023	-2.295	2.919	3.713

remains slightly higher than the DTU17, SS 32.1, and LSC models. Moreover, the effective wavelength of the RBF model reduces to <10 km, offering superior performance compared to the DTU17, SS 32.1, and LSC models. In the central-eastern Philippine Sea region, the LSC model maintains strong coherence with the shipborne data across the entire wavelength range. The coherence between the LSC and RBF models with shipborne data remains generally consistent in the medium-to-long wavelengths. However, below the cutoff wavelength (approximately 17 km), the LSC model exhibits higher coherence than the RBF model. As shown in Fig. 15(d)–(f), the effective wavelength of the RBF model is reduced from ~17 to <10 km, surpassing that of the SS 32.1 model. This improvement is attributed to the addition of shipborne data, which effectively compensates for the short-wavelength information.

Spectral analysis in the central-eastern Philippine Sea region shows that, for the HT91T233, HT94T314, and HT95-T343 survey lines, the power spectra of the LSC model and the shipborne data are generally consistent across the entire wavelength range. However, since these three survey lines have already been utilized in the fusion models, the power spectrum of the fusion models was again calculated using three external survey lines: MR0703, MR1301, and MR1303Leg2. Before calculating the power spectrum, Table IV presents the accuracy statistics for the fusion models across these three external survey lines. As indicated in Table III, the RBF model exhibits superior accuracy compared to the LSC model. Fig. 16 shows the coherence spectrum of the SS 32.1 model with the three external survey lines. Furthermore, the power spectra of the DTU17, SS 32.1, LSC, and RBF models, along with the shipborne data, were calculated and presented in Fig. 17. In

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

the short-wavelength range, the LSC and RBF models show better performance than the DTU17 and SS 32.1 models, with the RBF model surpassing the LSC model. For instance, on the MR0703 survey line [see Fig. 17(a)], below 2 km, the RBF model agrees more closely with shipborne data and significantly outperforms the LSC model. A comparison of Fig. 14(d)–(f) with Fig. 17 reveals that although the LSC model demonstrates considerable advantages on survey lines involved in the fusion models, it does not exhibit the same level of performance on external survey lines. In contrast, the RBF model demonstrates more stable performance and outperforms the LSC model on external survey lines. Moreover, in both Figs. 14 and 17, the RBF model effectively recovers the short-wavelength gravity signals.

V. Conclusion

This study develops an iterative RBF framework for fusing satellite altimetry and shipborne gravity data. The framework is applied to gravity field refinement experiments in the Ogasawara Islands and the central-eastern Philippine Sea regions. The iterative process effectively detects and removes gross errors while correcting systematic errors in the shipborne data, significantly enhancing data quality. The accuracy of the derived RBF model is notably improved with the iterative framework. In the Ogasawara Islands region, the standard deviation of the RBF model decreased from 6.297 to 2.951 mGal after the iterative process, resulting in an accuracy improvement of approximately 53.14%. Similarly, in the central-eastern Philippine Sea region, the standard deviation was reduced from 3.544 to 2.949 mGal, yielding an accuracy improvement of approximately 16.79%.

Both the LSC and RBF models effectively fuse altimetry and shipborne data, improving the accuracy of the regional marine gravity field model and complementing the altimetry signal in the short-wavelength range. For example, the RBF model primarily complements high-frequency signals within the 8–17 km wavelength range. Compared to the DTU17 and SS 32.1 models, the RBF model improves in accuracy by about 28.59% and 16.68%, respectively, in the Ogasawara Islands region. In the central-eastern Philippine Sea, the LSC model shows accuracy improvements of approximately 13.68% and 14.15% over DTU17 and SS 32.1, while the RBF model achieves improvements of 40.23% and 40.56%. Power spectrum and coherence analyses further demonstrate that in the Ogasawara Islands region, the LSC model matches shipborne data more closely in the 2–6 km wavelength range, with an effective wavelength reduced to approximately 14.5 km. In the central-eastern Philippine Sea region, the LSC model also maintains high coherence with the shipborne data in the short-wavelength range. In both regions, the RBF model consistently complements high-frequency signals within 8–17 km, reducing effective wavelength to <10 km, and surpassing both the DTU17 and SS 32.1 models.

The RBF model exhibits superior generalization capability compared to the LSC model, which is attributed to the local support properties of RBFs. In the Ogasawara Islands region, the RBF model consistently outperforms the LSC model when compared to the shipborne data used for model construction (see Table II) and external validation (see Table III). In the central-eastern Philippine Sea region, however, the LSC model shows higher accuracy than the RBF model when evaluated against the shipborne data involved in modeling, with a standard deviation of 0.365 mGal compared to 1.487 mGal for the RBF model. Moreover, spectral analysis along the three survey lines (HT91T233, HT94T314, and HT95T343) further confirms the better performance of the LSC model in this region. In external validation, the standard deviation of the LSC model increases to 4.259 mGal, whereas that of the RBF model is 2.949 mGal, indicating significantly higher accuracy of the latter. On the three external survey lines (MR0703, MR1301, and MR1303Leg2), the accuracy of the RBF model is also superior to that of the LSC model, and the power spectrum analysis shows that the RBF model outperforms the LSC model in the short-wavelength range. The Ogasawara Islands region, characterized by large geological structures and significant topographic variations, is less suitable for the LSC method. Conversely, in the central-eastern Philippine Sea region, the LSC model performs notably better on the shipborne data used in modeling, including both accuracy and spectral performance. It is noteworthy that the LSC model does not perform as effectively on external shipborne data. In contrast, the RBF model exhibits greater stability and outperforms the LSC model when applied to external data.

The findings of this study demonstrate the effectiveness of the iterative RBF framework in refining the regional marine gravity field through the fusion of satellite altimetry and shipborne data. Future research will focus on developing a global marine gravity field model by fusing global shipborne gravity and satellite altimetry gravity anomalies. Nonetheless, several aspects remain to be explored. For instance, although the RBF model has demonstrated significant improvements in complex topography areas, there is still potential for further enhancement in areas such as seamount chains and ridges. The ability of a single-depth, uniformly distributed RBF network to represent short-wavelength signals may be limited. A multiresolution RBF network would be better suited to adapt to local complex topography and data distribution. Nevertheless, RBF networks possess high flexibility, though the optimal determination of planar positions and depths lacks explicit regularity—being closely related to regional characteristics including area size, terrain features, data volume, and distribution patterns. Excessively shallow depths of RBFs may maintain higher modeling stability but risk overfitting, whereas overly deep configurations would compromise modeling stability. The spatial resolution of RBFs also impacts both model stability and accuracy: an excessive number of basis functions may lead to ill-conditioned normal equations, while insufficient quantities might degrade solution precision. This study determined appropriate combinations of the quantity and depth of RBFs through multiple experimental sets. However, this approach requires substantial computational experimentation and presents certain limitations for constructing global marine gravity field models. Some researchers have discussed issues related to the design of basis function networks [44], [45], but further research and analysis are required to improve

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

the flexibility, applicability, and stability of basis function networks.

ACKNOWLEDGMENT

The authors would like to thank the Supercomputing Center of Wuhan University for their computational support and assistance. They also acknowledge the marine gravity field models provided by the DTU, SIO, as well as the sea surface shipborne gravity data from the NCEI.

REFERENCES

[1] D. McAdoo and S. Laxon, "Antarctic tectonics: Constraints from an ERS-1 satellite marine gravity field," *Science*, vol. 276, no. 5312, pp. 556–561, Apr. 1997, doi: 10.1126/science.276.5312.556.

[2] J. D. Fairhead, C. M. Green, and M. E. Odegard, "Satellite-derived gravity having an impact on marine exploration," *Lead. Edge*, vol. 20, no. 8, pp. 873–876, Aug. 2001, doi: 10.1190/1.1487298.

[3] Y. Wu et al., "Seafloor topography modeling by fusing ICESat-2 LiDAR, echo sounding, and airborne and altimetric gravity data from spherical radial basis functions," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4201317, doi: 10.1109/TGRS.2024.3523893.

[4] V. Lieb, "Enhanced regional gravity field modeling from the combination of real data via MRR," Ph.D. dissertation, Dept. Civil Geo Environ. Eng., Tech. Univ. München, Munich, Germany, 2017.

[5] Q. Liu, M. Schmidt, and L. Sánchez, "Combination of different observation types through a multi-resolution representation of the regional gravity field using the pyramid algorithm and parameter estimation," *J. Geodesy*, vol. 96, no. 10, Oct. 2022, Art. no. 80, doi: 10.1007/s00190-022-01670-5.

[6] D. Bolkas, G. Fotopoulos, and A. Braun, "Comparison and fusion of satellite, airborne, and terrestrial gravity field data using wavelet decomposition," *J. Surveying Eng.*, vol. 142, no. 2, May 2016, Art. no. 04015010, doi: 10.1061/(asce)su.1943-5428.0000162.

[7] W. A. Heiskanen and H. Moritz, "The gravity field of the Earth," in *Physical Geodesy*, vol. 5, 1st ed., San Francisco, CA, USA: W. H. Freeman, 1967, ch. 2, pp. 57–60.

[8] M. Naeimi, "Inversion of satellite gravity data using spherical radial base functions," Ph.D. dissertation, Inst. Für Erdmessung, Leibniz Univ. Hannover, Hannover, Germany, 2013.

[9] Y. Wu, Z. Luo, W. Chen, and Y. Chen, "High-resolution regional gravity field recovery from Poisson wavelets using heterogeneous observational techniques," *Earth, Planets Space*, vol. 69, no. 1, Feb. 2017, Art. no. 34, doi: 10.1186/s40623-017-0618-2.

[10] R. Forsberg and C. C. Tscherning, "The use of height data in gravity field approximation by collocation," *J. Geophys. Res., Solid Earth*, vol. 86, no. B9, pp. 7843–7854, Sep. 1981, doi: 10.1029/jb086ib09p07843.

[11] C. C. Tscherning, F. Rubek, and R. Forsberg, "Combining airborne and ground gravity using collocation," in *Geodesy on the Move: Gravity, Geoid, Geodynamics and Antarctica*. Berlin, Germany: Springer, 1998, pp. 18–23, doi: 10.1007/978-3-642-72245-5_3.

[12] M. D. Buhmann, "Radial basis functions with compact support," in *Radial Basis Functions: Theory and Implementations*, vol. 3, 1st ed., Cambridge, U.K.: Cambridge Univ. Press, 2003, ch. 6, pp. 153–159.

[13] F. Barthelmes, "Local gravity field approximation by point masses with optimized positions," in *Proc. 6th Int. Symp. Geodesy Phys. Earth*, 1988, pp. 157–167.

[14] M. Vermeer, "Mass point geopotential modelling using fast spectral techniques; historical overview, toolbox description, numerical experiment," *Manuscripta Geodaetica*, vol. 20, no. 5, pp. 362–378, Jun. 1995, doi: 10.1007/bf03655470.

[15] A. N. Marchenko, F. Barthelmes, U. Meyer, and P. Schwintzer, "Regional geoid determination: An application to airborne gravity data in the Skagerrak," GeoForschZentrum Potsdam, Potsdam, Germany, Tech. Rep. STR01/07, 2001.

[16] A. N. Marchenko, "Regional gravity field model from satellite altimetry, gravimetry, and GPS-leveling data in the black sea area," *Bolletino Di Geodesia E Scienze Affini*, vol. 62, no. 4, pp. 246–259, 2003.

[17] R. Tenzer and R. Klees, "The choice of the spherical radial basis functions in local gravity field modeling," *Studia Geophysica Et Geodaetica*, vol. 52, no. 3, pp. 287–304, Sep. 2008, doi: 10.1007/s11200-008-0022-2.

[18] T. Wittwer, "Regional gravity field modelling with radial basis functions," Ph.D. dissertation, Dept. Earth Obs. Space Syst., Delft Univ. Technol., Delft, The Netherlands, 2009.

[19] A. Chambodut, I. Panet, M. Mandea, M. Diament, M. Holschneider, and O. Jamet, "Wavelet frames: An alternative to spherical harmonic representation of potential fields," *Geophys. J. Int.*, vol. 163, no. 3, pp. 875–899, Dec. 2005, doi: 10.1111/j.1365-246x.2005.02754.x.

[20] M. Holschneider and I. Iglewska-Nowak, "Poisson wavelets on the sphere," *J. Fourier Anal. Appl.*, vol. 13, no. 4, pp. 405–419, May 2007, doi: 10.1007/s00041-006-6909-9.

[21] Y. Wu, H. Zhou, B. Zhong, and Z. Luo, "Regional gravity field recovery using the GOCE gravity gradient tensor and heterogeneous gravimetry and altimetry data," *J. Geophys. Res., Solid Earth*, vol. 122, no. 8, pp. 6928–6952, Aug. 2017, doi: 10.1002/2017jb014196.

[22] R. B. Blackman and J. W. Tukey, "The measurement of power spectra from the point of view of communications engineering—Part II," *Bell Syst. Tech. J.*, vol. 37, no. 2, pp. 485–569, Mar. 1958, doi: 10.1002/j.1538-7305.1958.tb01530.x.

[23] M. Schmidt, "Gravity field determination using multiresolution techniques," in *Proc. 2nd Int. GOCE User Workshop*, vol. 569, Frascati, Italy, 2004, p. 35.

[24] M. Schmidt et al., "Regional gravity modeling in terms of spherical base functions," *J. Geodesy*, vol. 81, no. 1, pp. 17–38, Oct. 2006, doi: 10.1007/s00190-006-0101-5.

[25] B. Bucha, A. Bezděk, J. Sebera, and J. Janák, "Global and regional gravity field determination from GOCE kinematic orbit by means of spherical radial basis functions," *Surv. Geophys.*, vol. 36, no. 6, pp. 773–801, Oct. 2015, doi: 10.1007/s10712-015-9344-0.

[26] K. Bentel, M. Schmidt, and C. Rolstad Denby, "Artifacts in regional gravity representations with spherical radial basis functions," *J. Geodetic Sci.*, vol. 3, no. 3, pp. 173–187, Jan. 2013, doi: 10.2478/jogs-2013-0029.

[27] R. Klees, R. Tenzer, I. Prutkin, and T. Wittwer, "A data-driven approach to local gravity field modelling using spherical radial basis functions," *J. Geodesy*, vol. 82, no. 8, pp. 457–471, Feb. 2008, doi: 10.1007/s00190-007-0196-3.

[28] X. Li, "Using radial basis functions in airborne gravimetry for local geoid improvement," *J. Geodesy*, vol. 92, no. 5, pp. 471–485, May 2018, doi: 10.1007/s00190-017-1074-2.

[29] Q. Liu, M. Schmidt, L. Sánchez, and M. Willberg, "Regional gravity field refinement for (quasi-) geoid determination based on spherical radial basis functions in Colorado," *J. Geodesy*, vol. 94, no. 10, Oct. 2020, Art. no. 99, doi: 10.1007/s00190-020-01431-2.

[30] V. Lieb, M. Schmidt, D. Dettmering, and K. Börger, "Combination of various observation techniques for regional modeling of the gravity field," *J. Geophys. Res., Solid Earth*, vol. 121, no. 5, pp. 3825–3845, May 2016, doi: 10.1002/2015jb012586.

[31] Y. Wu, A. Abulaitijiang, O. B. Andersen, X. He, Z. Luo, and H. Wang, "Refinement of mean dynamic topography over island areas using airborne gravimetry and satellite altimetry data in the Northwestern South China Sea," *J. Geophys. Res., Solid Earth*, vol. 126, no. 8, Aug. 2021, Art. no. e2021JB021805, doi: 10.1029/2021jb021805.

[32] Z. Ma, "Gravity field modeling in mountainous areas based on band-limited SRBFs," *J. Geodesy*, vol. 98, no. 5, May 2024, Art. no. 41, doi: 10.1007/s00190-024-01852-3.

[33] A. N. Tikhonov, "Solution of incorrectly formulated problems and the regularization method," *Sov. Math. Dokl.*, vol. 4, pp. 1035–1038, 1963.

[34] X. Xu, W. Jiang, X. Zhang, X. Zhou, Y. Ding, and G. Zhu, "Ability of recovering the global gravity field of a new satellite gravimetry system," *Chin. J. Geophys.*, vol. 61, no. 6, pp. 2227–2236, Jun. 2018, doi: 10.6038/cjg2018L0270.

[35] Y. Zhao, J. Li, X. Xu, Y. Su, H. Zong, and H. Wei, "Determination of static gravity field model by using satellite data of GOCE and GRACE," *Chin. J. Geophys.*, vol. 66, no. 6, pp. 2322–2336, Jun. 2023, doi: 10.6038/cjg2022Q0615.

[36] O. Andersen and P. Knudsen, "The DTU17 global marine gravity field: First validation results," in *Proc. Int. Rev. Workshop Satell. Altim. Cal/Val*, Chania, Greece, 2019, pp. 83–87, doi: 10.1007/1345_2019_65.

[37] R. Reuter, "Über integralformeln der einheitssphäre und harmonische splinefunktionen," Veröff. Geod. Inst., Aachen, Germany, Tech. Rep. 33, 1982.

[38] A. Eicker, "Gravity field refinement by radial basis functions from in-situ satellite data," Ph.D. dissertation, Fac. of Agricult., Univ. Bonn, Bonn, Germany, 2008.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.

[39] V. Lieb, J. Bouman, D. Dettmering, M. Fuchs, and M. Schmidt, "Combination of GOCE gravity gradients in regional gravity field modelling using radial basis functions," in *Proc. 7th Hotine-Marussi Symp. Math. Geodesy. IAG Symposia*, Rome, Italy, 2015, pp. 101–108, doi: 10.1007/1345_2015_71.
[40] R. Forsberg, "A new covariance model for inertial gravimetry and gradiometry," *J. Geophys. Res., Solid Earth*, vol. 92, no. B2, pp. 1305–1310, Feb. 1987, doi: 10.1029/jb092ib02p01305.
[41] P. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Trans. Audio Electroacoustics*, vol. AE-15, no. 2, pp. 70–73, Jun. 1967.
[42] K. M. Marks and W. H. F. Smith, "Detecting small seamounts in AltiKa repeat cycle data," *Mar. Geophys. Res.*, vol. 37, no. 4, pp. 349–359, Nov. 2016, doi: 10.1007/s11001-016-9293-0.
[43] X. Chen, X. Kong, R. Zhou, and S. Zhang, "Fusion of altimetry-derived model and ship-borne data in preparation of high-resolution marine gravity determination," *Geophys. J. Int.*, vol. 236, no. 3, pp. 1262–1274, Jan. 2024, doi: 10.1093/gji/ggad471.
[44] H. Mahbuby, A. Safari, and I. Foroughi, "Local gravity field modeling using spherical radial basis functions and a genetic algorithm," *Comp. Rendus. Géoscience*, vol. 349, no. 3, pp. 106–113, May 2017, doi: 10.1016/j.crte.2017.03.001.
[45] R. Ulug and M. O. Karslıoglu, "A new data-adaptive network design methodology based on the k-means clustering and modified ISODATA algorithm for regional gravity field modeling via spherical radial basis functions," *J. Geodesy*, vol. 96, no. 11, Nov. 2022, Art. no. 91, doi: 10.1007/s00190-022-01681-2.



Wenqi Lin was born in Gansu, China. He received the M.S. degree in geodesy from Chinese Academy of Surveying and Mapping, Beijing, China, in 2022. He is currently pursuing the Ph.D. degree with Wuhan University, Wuhan, China.

His research interests include multisource gravity data fusion and regional gravity field modeling.



Jiancheng Li received the B.Eng. degree in engineering surveying and the M.S. and Ph.D. degrees in geodesy from Wuhan University, Wuhan, China, in 1987, 1990, and 1993, respectively.

He is currently a Professor with the School of Geodesy and Geomatics, Wuhan University. He has authored or co-authored over 120 articles. His main research interests include Earth's gravity field theory, geoid refinement, gravity inversion, and time-varying gravity field modeling and applications.



Xinyu Xu received the B.Eng. degree in engineering surveying and the M.S. and Ph.D. degrees in geodesy from Wuhan University, Wuhan, China, in 2001, 2004, and 2008, respectively.

He is currently a Professor with the School of Geodesy and Geomatics, Wuhan University. He has authored over 20 articles. His main research interests are the theoretical methods and applications for fine global gravity field modeling.



Yongqi Zhao received the M.S. and Ph.D. degrees in geodesy from Wuhan University, Wuhan, China, in 2016 and 2023, respectively.

He is currently a Post-Doctoral Researcher with Wuhan University. His research interests include satellite data processing from GOCE/GRACE and global gravity field modeling.



Mao Zhou received the M.S. degree in geodesy from Wuhan University, Wuhan, China, in 2021, where he is currently pursuing the Ph.D. degree.

His research interests include satellite altimetry data processing, marine gravity field recovery, and bathymetry prediction from altimetry.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:04:31 UTC from IEEE Xplore. Restrictions apply.